Service and Category Management System

NAILVERSE

Not just NAILS.. IT's the WHOLE MOOD

Online Nail Art & Nail Beauty Online Platform

WEEK - 2

Scenario

Nailverse wants to manage information about the nail services it provides and the categories those services fall under. Each service belongs to a specific category (e.g. Manicure, Pedicure, Nail Art, Extensions), and the system should maintain service details such as name, price, duration, and slot availability. This continues directly from the Service and Category attributes outlined in the Week 1 entity list.

1. Design Service and Category Tables

Category Table:

```
SQL> create table Category(Category_ID VARCHAR2(7),
  2  Category_Name VARCHAR2(30),
  3  Description VARCHAR(100));

Table created.
```

Service Table:

```
SQL> create table service(Service_ID VARCHAR2(7),
  2  Service_Name VARCHAR2(50),
  3  Category_ID VARCHAR2(7),
  4  Price NUMBER(8,2),
  5  Duration_Minutes NUMBER,
  6  Slot_Available NUMBER);

Table created.
```

The Category and Service tables have now been created.

Next, we need to identify the relationship between the tables and apply the required constraints.

2. Define Relationships and Constraints

Relationships:

A single Category can contain multiple Services.

The relationship between Category and Service is One to many (1 -> *)

After identifying the relationship, we need to determine which key will enforce it.
Before proceeding, we need to understand KEY constraints
.

- PRIMARY KEY: The column or field that uniquely identifies a row or tuple.

Which field can serve as the Primary Key in these two tables?
In the Category Table, Category_ID is a unique value used to identify each category.
Similarly, in the Service Table, Service_ID is a unique column.

- FOREIGN KEY: The key that connects two tables.

Therefore, Category_ID acts as a Foreign Key in the Service Table.
We can also consider other constraints such as NOT NULL, CHECK and UNIQUE.

- NOT NULL: Ensures that a column cannot contain an empty (null) value. For example, a Service name or Category name cannot be empty. Each category or service must have a name.

- CHECK: Checks that the values in a column meet a specified condition. For example, the Price of a Service should always be a positive number, and Slots_Available cannot be negative.

- UNIQUE: Ensures that a column does not contain repeated values. For example, if we apply a UNIQUE constraint to Category_Name, no two categories can have the same name.

Defining Relationship and Creating Constraints for Service and Category:
Primary Key:

```
SQL> ALTER TABLE Category
  2  ADD CONSTRAINT PK_Category PRIMARY KEY (Category_ID);

Table altered.
SQL> ALTER TABLE service
  2  ADD CONSTRAINT PK_service PRIMARY KEY (service_ID);

Table altered.
```

Creating Relationship using FOREIGN KEY:

```
SQL> ALTER TABLE service
  2  ADD CONSTRAINT FK_service_Category
  3  FOREIGN KEY(Category_ID) REFERENCES Category(Category_ID);

Table altered.
```

Adding Constraints:

NOT NULL:

```
SQL> ALTER TABLE Category
  2  MODIFY Category_Name VARCHAR2(30) NOT NULL;

Table altered.


SQL> ALTER TABLE Service
  2  MODIFY ServIce_Name VARCHAR2(50)  NOT NULL;

Table altered.
```

CHECK:

```
SQL> ALTER TABLE Service
  2  ADD CONSTRAINT CHK_Price CHECK (Price > 0);

Table altered.
```

```
SQL> ALTER TABLE Service
  2  ADD CONSTRAINT CHEK_Slots CHECK (Slot_Available >= 0);

Table altered.
```

UNIQUE:

```
SQL> ALTER TABLE Category
  2  ADD CONSTRAINT UQ_Category_Name UNIQUE (Category_Name);

Table altered.
```

3. Insert Values

Category Table:

```
SQL> INSERT INTO Category
  2  VALUES (
  3  '&Category_ID',
  4  '&Category_Name',
  5  '&Description');
Enter value for category_id: CAT001
old   3: '&Category_ID',
new   3: 'CAT001',
Enter value for category_name: MANICURE
old   4: '&Category_Name',
new   4: 'MANICURE',
Enter value for description: HAND AND NAIL CARE SERVICE
old   5: '&Description')
new   5: 'HAND AND NAIL CARE SERVICE')

1 row created.

SQL> /
Enter value for category_id: CAT002
old   3: '&Category_ID',
new   3: 'CAT002',
Enter value for category_name: PEDICURE
old   4: '&Category_Name',
new   4: 'PEDICURE',
Enter value for description: FOOT AND NAIL CARE SERVICES
old   5: '&Description')
new   5: 'FOOT AND NAIL CARE SERVICES')

1 row created.
```

Like this, insert the remaining categories: Nail Art, Extensions.

Service Table:

```
SQL> INSERT INTO service
  2  VALUES(
  3  '&Service_ID',
  4  '&Service_Name',
  5  '&Category_ID',
  6  &Price,
  7  '&Duration_Minutes',
  8  '&Slots_Available');
Enter value for service_id: SRV001
old   3: '&Service_ID',
new   3: 'SRV001',
Enter value for service_name: CLASSIC MAINCURE
old   4: '&Service_Name',
new   4: 'CLASSIC MAINCURE',
Enter value for category_id: CAT001
old   5: '&Category_ID',
new   5: 'CAT001',
Enter value for price: 499
old   6: &Price,
new   6: 499,
Enter value for duration_minutes: 50
old   7: '&Duration_Minutes',
new   7: '50',
Enter value for slots_available: 20
old   8: '&Slots_Available')
new   8: '20')

1 row created.
```

Like this insert few more rows.

After Insertion – Category Table:

```
SQL> SELECT * FROM Category;

CATEGOR CATEGORY_NAME
------- ------------------------------
DESCRIPTION
--------------------------------------------------------------------------------
CAT001  MANICURE
HAND AND NAIL CARE SERVICE

CAT002  PEDICURE
FOOT AND NAIL CARE SERVICES

CAT003  NAIL EXTENSIONS
ACRYLIC AND GEL NAIL EXTENSIONS


CATEGOR CATEGORY_NAME
------- ------------------------------
DESCRIPTION
--------------------------------------------------------------------------------
CAT004  NAIL ART
CREATIVE NAIL DESIGNS AND DECORATION

CAT005  SPA CARE
RELAXING HAND AND FOOT SPA SERVICES
```

After Insertion – Service Table:

```
SQL> select * from service;

SERVICE SERVICE_NAME                                       CATEGOR      PRICE
------- -------------------------------------------------- ------- ----------
DURATION_MINUTES SLOT_AVAILABLE
---------------- --------------
SRV001  CLASSIC MAINCURE                                   CAT001         499
              50             20

SRV002  CLASSIC PEDICURE                                   CAT002         559
              45             20

SRV003  NAIL EXTENSIONS                                    CAT003         459
              50             15


SERVICE SERVICE_NAME                                       CATEGOR      PRICE
------- -------------------------------------------------- ------- ----------
DURATION_MINUTES SLOT_AVAILABLE
---------------- --------------
SRV004  CHROME NAIL ART                                    CAT004         999
              45             25

SRV005  SPA PEDICURE                                       CAT005         799
              90             10
```

CRUD Operations:

Insert Record:

```
SQL> INSERT INTO SERVICE
  2      (SERVICE_ID, SERVICE_NAME, CATEGORY_ID, PRICE, DURATION
_MINUTES)
  3  VALUES
  4      ('SRV006', 'Chrome Nail Art', 'CAT003', 1299, 80);

1 row created.
```

Update Price:

```
SQL> UPDATE service SET Price = 650 WHERE Service_Name = 'CLASSI
C PEDICURE';

1 row updated.
```

5. Generate Category-wise Service Reports

Write SQL queries to:

- Display all services under each category.

```
SQL> SELECT c.Category_Name,
  2         s.Service_Name,
  3         s.Price,
  4         s.Duration_Minutes
  5  FROM Service s
  6  JOIN Category c
  7  ON s.Category_ID = c.Category_ID;

CATEGORY_NAME
------------------------------
SERVICE_NAME                                       PRICE DURATION_MINUTES
-------------------------------------------------- ---------- ----------------
MANICURE
CLASSIC MAINCURE                                     499       50

PEDICURE
CLASSIC PEDICURE                                     650       45

NAIL EXTENSIONS
NAIL EXTENSIONS                                      459       50


CATEGORY_NAME
------------------------------
SERVICE_NAME                                       PRICE DURATION_MINUTES
-------------------------------------------------- ---------- ----------------
NAIL ART
CHROME NAIL ART                                      999       45

SPA CARE
SPA PEDICURE                                         799       90

NAIL EXTENSIONS
Chrome Nail Art                                     1299       80


6 rows selected.
```

- Count number of services in each category.

```
SQL> SELECT c.Category_Name, COUNT(s.Service_ID) AS Service_Count FROM Serv
ice s JOIN Category c ON s.Category_ID = c.Category_ID GROUP BY c.Category_
Name;

CATEGORY_NAME                 SERVICE_COUNT
------------------------------ -------------
MANICURE                                   1
PEDICURE                                   1
NAIL ART                                   1
NAIL EXTENSIONS                            2
SPA CARE                                   1
```

- Find highest-priced service in each category.

```
SQL> SELECT c.Category_Name, s.Service_Name, s.Price AS Highest_Price FROM
Service s JOIN Category c ON s.Category_ID = c.Category_ID WHERE s.Price =
(SELECT MAX(s2.Price) FROM Service s2 WHERE s2.Category_ID = s.Category_ID)
;

CATEGORY_NAME
------------------------------
SERVICE_NAME                                       HIGHEST_PRICE
-------------------------------------------------- -------------
MANICURE
CLASSIC MAINCURE                                             499

PEDICURE
CLASSIC PEDICURE                                             650

NAIL ART
CHROME NAIL ART                                              999


CATEGORY_NAME
------------------------------
SERVICE_NAME                                       HIGHEST_PRICE
-------------------------------------------------- -------------
SPA CARE
SPA PEDICURE                                                 799

NAIL EXTENSIONS
Chrome Nail Art                                             1299
```

- Display categories having more than 2 services.

```
SQL> SELECT c.Category_Name,
  2         COUNT(s.Service_ID) AS Total_Services
  3  FROM Category c
  4  LEFT JOIN Service s
  5  ON c.Category_ID = s.Category_ID
  6  GROUP BY c.Category_Name;

CATEGORY_NAME                  TOTAL_SERVICES
------------------------------ --------------
MANICURE                                    1
PEDICURE                                    1
NAIL ART                                    1
NAIL EXTENSIONS                             2
SPA CARE                                    1
```

- Find average service price category-wise.

```
SQL> SELECT c.Category_Name,
  2         AVG(s.Price) AS Avg_Price
  3  FROM Service s
  4  JOIN Category c
  5  ON s.Category_ID = c.Category_ID
  6  GROUP BY c.Category_Name;

CATEGORY_NAME                  AVG_PRICE
------------------------------ ----------
MANICURE                              499
PEDICURE                              650
NAIL ART                              999
NAIL EXTENSIONS                       879
SPA CARE                              799
```